

Introduction to AI

Intelligent agents

What is Artificial intelligence (AI)?

AI: notion introduced by John McCarthy in 1956:

"the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but *AI does not have to confine itself to methods that are biologically observable.*"

There are two versions of AI:

- 1 Acting like a human
- 2 Acting rationally (or "doing the right thing")

1. Acting like a human

The Turing Test and Chatterbots

Turing test, a.k.a. *imitation game* (Turing, 1950): a thought experiment intended to decide if a machine can think:

A human interrogator communicates with both a human and a machine (computer) through text, aiming to determine which is the machine. If the interrogator cannot reliably distinguish the machine from the human, the machine is considered to have passed the test: it can think.

Nowadays, the **total Turing test** requires human-like interaction with objects and people in the real world $\Rightarrow$ the following capabilities:

natural language processing to communicate successfully in a human language

knowledge representation to store what it knows or hears

automated reasoning to answer questions and draw more conclusions

machine learning to adapt to new circumstances and to detect and extrapolate new patterns

computer vision and speech recognition to perceive the world

robotics to manipulate objects and move

1. Acting like a human

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Is the Turing test still relevant?

Note: The large language models that work with deep learning, such as ChatGPT, pass the Turing test with ease.

1. Acting like a human

Thinking humanly

The study of human-like intelligence is related to psychology, involving observations about human behavior and thought processes.

The cognitive modeling approach of human thinking is based on

introspection: trying to catch our own thoughts

psychological experiments: observing a person in action

brain imaging: observing the brain in action

2. Acting rationally

Rational agents. Thinking rationally

The **standard model** of AI is concerned with the study and construction of **rational agents**.

- an **AI agent** should operate autonomously, perceive their environment, persists over a prolonged time period, adapt to change, and create and pursue goals.
- a **rational AI agent** acts to achieve the **best outcome** or, when there is uncertainty, the **best expected outcome**.

Rational thinking is the study of the laws of thought that govern the processes of correct human reasoning: *how to draw correct conclusions when given correct premises*.

- **Logic** is the field of study of rational thinking.
- Usually, logic requires knowledge of the world that is **certain**. Often, we must reason with **uncertain** information
⇒ **probability theory** allows reasoning with uncertain information.

Rational agents

Remarks

Rational agent = **act rationally** = achieve the **best outcome**.

Simply put: *do the right thing.*

Knowledge representation and reasoning enable agents to act rationally $\Rightarrow$ **rational thinking** is one of several possible mechanisms to achieve rational action.

$\Rightarrow$ the **logician approach** to AI = design **logical agents**, which use knowledge representation and logic to decide what actions to take.

Rational thinking is not necessary in the construction of a rational agent.

- Rationality can be defined in a rigorous mathematical way: minimise the cost function of a controller (Control Theory); maximise a sum of rewards (Operations Research); minimise a loss function (Statistics); etc.
- **Perfect rationality** (=always taking the optimal action) is not feasible in complex environments $\Rightarrow$ construction of agents with **Limited rationality** = acting appropriately when there is not enough time to do all computations one may like.

History of AI

Milestones (1)

- 1931: The Austrian Kurt Gödel shows that in first-order predicate logic all true statements are derivable [Göd31a]. In higher-order logics, on the other hand, there are true statements that are unprovable [Göd31b]. (In [Göd31b] Gödel showed that predicate logic extended with the axioms of arithmetic is incomplete)
- 1937: Turing points out the limits of intelligent machines with the halting problem
- 1943: McCulloch and Pitts model *neural networks* and make the connection to propositional logic
- 1950: Alan Turing defines machine intelligence with the Turing test and writes about learning machines and genetic algorithms [Tur50]
- 1951: Marvin Minsky develops a neural network machine. With 3000 vacuum tubes he simulates 40 neurons
- 1955: Arthur Samuel (IBM) builds a learning checkers program that plays better than its developer
- 1956: McCarthy organizes a conference in Dartmouth College. Here the name *Artificial Intelligence* was first introduced
- 1956:1 Newell and Simon of Carnegie Mellon University (CMU) present the Logic Theorist, the first symbol-processing computer program

History of AI

Milestones (2)

- 1958: McCarthy invents at MIT (Massachusetts Institute of Technology) the high-level language LISP. He writes programs that are capable of modifying themselves
- 1959: Gelernter (IBM) builds the Geometry Theorem Prover
- 1961: The General Problem Solver (GPS) by Newell and Simon imitates human thought [NS61]
- 1963: McCarthy founds the AI Lab at Stanford University
- 1965: Robinson invents the resolution calculus for predicate logic [Rob65]
- 1966: Weizenbaum's program Eliza carries out dialog with people in natural language [Wei66]
- 1969: Minsky and Papert show in their book *Perceptrons* that the perceptron, a very simple neural network, can only represent linear functions
- 1972: A. Colmerauer invents the logic programming language PROLOG
- 1976: Shortliffe and Buchanan develop MYCIN, an expert system for diagnosis of infectious diseases, which is capable of dealing with uncertainty
- 1981: Japan begins the "Fifth Generation Project" with the goal of building a powerful PROLOG machine
- 1982: R1, the expert system for configuring computers, saves Digital Equipment Corporation 40 million dollars per year

History of AI

Milestones (3)

- 1986: Renaissance of neural networks through, among others, Rumelhart, Hinton, and Sejnowski [RM86]. The system Nettetalk learns to read texts aloud [SR86]
- 1988: Pearl [Pea88], Cheeseman [Che85], Whittaker, Spiegelhalter bring probability theory into AI with *Bayesian networks*. Multi-agent systems become popular
- 1992: Tesauros TD-gammon program demonstrates the advantages of reinforcement learning
- 1993: Worldwide *RoboCup* initiative to build soccer-playing autonomous robots
- 1995: From statistical learning theory, Vapnik develops support vector machines, which are very important today
- 1997: IBM's chess computer Deep Blue defeats the chess world champion Gary Kasparov
- 1997: First international *RoboCup* competition in Japan
- 2003: The robots in *RoboCup* demonstrate impressively what AI and robotics are capable of achieving
- 2009: First Google self-driving car drives on the California freeway

History of AI

Milestones (4)

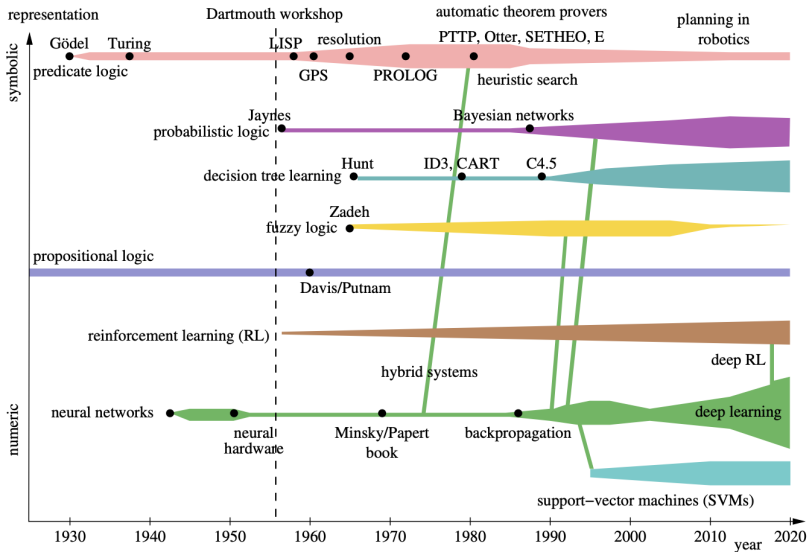
- 2010: Autonomous robots begin to improve their behavior through learning
- 2011: IBM's "Watson" beats two human champions on the television game show "Jeopardy!". Watson understands natural language and can answer difficult questions very quickly
- 2015:
 - Daimler premieres the first autonomous truck on the Autobahn
 - Google self-driving cars have driven over one million miles and operate within cities
 - Deep learning enables very good image classification
 - Paintings in the style of the Old Masters can be automatically generated with deep learning. AI becomes creative
 - Generative Adversarial Networks (GANs) can generate and manipulate photos and even videos. GANs can create images from nothing more than short descriptive texts

History of AI

Milestones (5)

- 2016: The Go program AlphaGo by Google DeepMind beats the European champion 5:0 in January and Korean Lee Sedol, one of the world's best Go players, 4:1 in March. Deep learning techniques applied to pattern recognition, as well as reinforcement learning and Monte Carlo tree search lead to this success
- 2017: AlphaGoZero learns to play Go on an even higher level by playing against itself using deep learning, reinforcement learning, and Monte Carlo tree search. Humans have no chance against AlphaGoZero. This approach is generalized in AlphaZero, which achieved superhuman chess abilities in 24 hours of training against itself
- 2020: The GPT3 language model trained on Wikipedia and many books by the company OpenAI can conduct academic conversations and passes the Turing Test

History of various AI areas



The width of the bars indicates prevalence of the method's use.

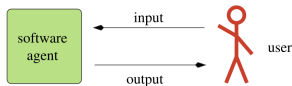
AI: what did we learn so far?

- The history of AI had cycles of success, misplaced optimism, and cutbacks in enthusiasm and funding. There have been cycles of introducing new, creative approaches and systematic refinements of the best ones.
- AI has matured considerably compared to 60 years ago, both theoretically and methodologically. The field of AI progressed
 - from Boolean logic to probabilistic reasoning
 - from hand-crafted knowledge to machine learning from data
- AI systems have many applications in the real world $\Rightarrow$ what are the risks and ethical consequences of using it?
- **Longer term concern:** super-intelligent AI systems may evolve in unpredictable ways $\Rightarrow$ how to keep control over them?

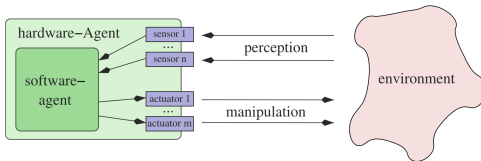
Agents and environments

The central concern of AI is the study and construction of **rational agents**.

- According to [RN21], an **agent** is anything that can **perceive** its **environment** through sensors and **act** upon the environment through **actuators**.
- In classical computer science, **software agents** (or **softbots**) are primarily employed. They consist of a program that calculates a result from user input.



- Robotics deals with **hardware agents** (*a.k.a.* autonomous robots) with sensors and actuators at their disposal



Every agent implements an **agent function** which specifies what action to take in response to the sequence of inputs received so far from its sensors.

Agents

The agent function. Rationality

The set of input sequences is **infinite** because they can get arbitrarily long.

- ⇒ the agent function can not be represented as an input-output table
- ⇒ AI is concerned to design **agent programs** which implements agent functions

A **rational agent** must do the **right thing**. The notion of *doing the right thing* can be described mathematically by a **performance measure**, which should be designed

according to what one actually wants to be achieved in the environment, rather than according to how one thinks the agent should behave.

Rationality depends on 4 things:

- 1 The **performance** (P) measure that defines the criterion of success.
- 2 The agent's prior knowledge of the **environment** (E)
- 3 The **actions** (A) that the agents can perform
- 4 The agent's percept **sequence** (S) to date.

⇒ the PEAS description of a rational agent, *a.k.a.* **task environment**:

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Intelligent agents

Examples of agent types and their PEAS descriptions

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Task environments

The construction of a rational agent starts from the description of a task environment.

Task environments can be classified along many dimensions:

- **Fully observable** vs. **partially observable**
- **Single agent** vs. **multiagent**
- **Deterministic** vs. **nondeterministic**
 - ▶ **Stochastic** agents are nondeterministic agents where we know the probabilities of different situations.
- ▶ **Episodic** vs. **sequential**
- **Static** vs. **dynamic**
 - ▶ does the environment change while an agent is deliberating?
- **Discrete** vs. **continuous**
 - ▶ depends on how *time* is handled
- **Known** vs. **unknown**
 - ▶ depends on the agent's (or designer's) state of knowledge about the "laws of physics" of the environment.

Remark: The hardest case is partially observable, multiagent, nondeterministic, sequential, dynamic, continuous, and unknown task environment.

Task environments and their characteristics

Examples

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Remark: The code repository associated with book [NR21] (aima.cs.berkeley.edu) includes multiple environment implementations, together with a general-purpose environment simulator for evaluating an agent's performance.

Structure of an agent

An agent program implements an agent function – the mapping from percepts to actions

- ▶ it will run on some sort of computing device with physical sensors and actuators – we call this the **agent architecture**

$$\text{agent} = \text{architecture} + \text{program}$$

- ▶ **Challenge for AI:** how to write programs that, to the extent possible, produce rational behavior from a small program rather than from a vast lookup table.

Example: Newton's method (5-line program) to compute $\sqrt{x}$. *Can AI do for general intelligent behaviour what Newton did for square roots?*

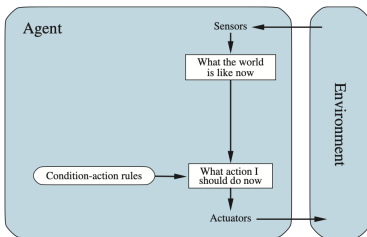
There are four basic kinds of agent programs that embody the principles underlying *almost* all intelligent systems:

- 1 Simple reflex agents
- 2 Model-based reflex agents
- 3 Goal-based agents
- 4 Utility-based agents

1. Simple reflex agents

They decide what action to take based only of the current percept.

- They work well only if the correct decision can be made on the basis of just the current percept – that is, only if the environment is fully observable.



function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
persistent: *rules*, a set of condition–action rules

state ← INTERPRET-INPUT(*percept*)

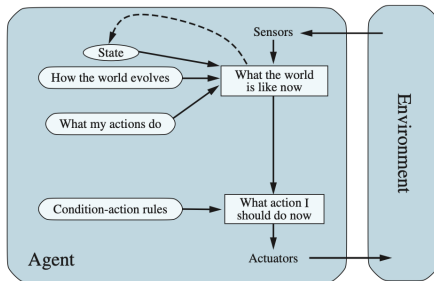
rule ← RULE-MATCH(*state*, *rules*)

action ← *rule*.ACTION

return *action*

2. Model-based agents

They store in an internal state a representation of the (part of the world) perceived with their sensors.



function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action

persistent: *state*, the agent's current conception of the world state

transition_model, a description of how the next state depends on
the current state and action

sensor_model, a description of how the current world state is reflected
in the agent's percepts

rules, a set of condition–action rules

action, the most recent action, initially none

state ← UPDATE-STATE(*state*, *action*, *percept*, *transition_model*, *sensor_model*)

rule ← RULE-MATCH(*state*, *rules*)

action ← *rule*.ACTION

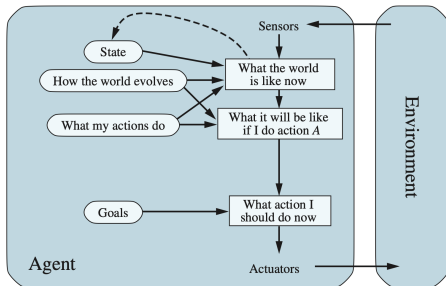
return *action*

3. Goal-based agents

They are designed to reach a **goal** (e.g., a taxi driver must reach a particular destination).

- **Search** and **planning** are the subfields of AI devoted to finding action sequences that achieve the agent's goals.
- Decision making of this kind involves consideration of the **future**; it is fundamentally different from the condition-action rules described earlier, which involve consideration of the **past**

Example of a model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.



4. Utility-based agents

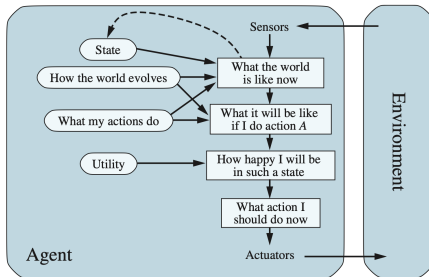
Goals are often not enough to generate high-quality behaviour. Utility-based agents use an **utility function** to measure the **utility** of different possible world states, and choose the one that maximises the utility.

- The utility measure of an action outcomes is a number.

Examples: how should we proceed when

- 1 we must choose between conflicting goals (e.g., speed vs safety)?
A: choose the one which is more useful to us?
- 2 we must make a choice among many goals, none of which can be achieved with certainty?
A: choose the one which maximises the **expected utility** of the action outcomes, that is, the utility the agent expects to derive, on average, given the probabilities and utilities of each outcome.

The structure of a model-based, utility-based agent is



More about utility-based agents

An alternative: learning agents

- Choosing the utility-maximizing course of action is a difficult task, requiring sophisticated algorithms that are difficult to hard-code.
- The complexity of some algorithms is too high $\Rightarrow$ they are practically useless.
- The **correct** specification of a utility function is a difficult task.

In 1950, A. Turing observed that programming intelligent machines by hand can require too much work, and concluded "*Some more expeditious method seems desirable.*"

- He proposed to build **learning machines** and then to **teach** them.
- In many areas of AI, learning became the preferred method for creating state-of-the-art systems.
- Any type of agent (model-based, goal-based, utility-based, etc.) can be built as a learning agent.
- Learning has an important advantage: it allows the agent to operate in initially unknown environments and to become more competent than its initial knowledge alone might allow.

References

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